

Smart Traffic Lights Using Local AI: An Intelligent Traffic Management System Using Real-Time Data and Adaptive Signal Control

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Abstract—Traffic congestion is a major challenge in modern urban transportation systems. Conventional fixed-time traffic signals cannot always respond effectively to rapidly changing traffic conditions, and existing adaptive systems do not automatically clear a path for a speeding ambulance. This paper presents a Smart Traffic framework that uses real-time traffic information processed by local AI for adaptive traffic signal control. The framework considers vehicle count, traffic density, queue length, vehicle speed, and waiting time as major traffic parameters, and integrates traffic sensing, data processing, intelligent decision making, and adaptive signal control. Mathematical formulations are given for traffic density, traffic flow, waiting time, and signal optimization. The framework can be extended using machine learning, reinforcement learning, Internet of Things, and vehicular communication technologies.

Index Terms—Smart Traffic, Local AI, Intelligent Transportation Systems, Traffic Signal Control, Reinforcement Learning, Internet of Things, Emergency Vehicle Priority

I. INTRODUCTION

Rapid urbanization and increasing vehicle ownership have created significant challenges for modern transportation systems. Traffic congestion increases travel time, fuel consumption, and vehicle emissions, and leads to inefficient use of road infrastructure.

Intelligent Transportation Systems (ITS) integrate sensing, communication, computation, and control technologies to improve transportation operations. Traditional traffic signals generally operate on predefined timing schedules and may not respond efficiently when traffic demand changes rapidly.

Reinforcement learning allows a traffic controller to learn signal-control policies from observed traffic conditions, and deep reinforcement learning for traffic signal control has been reviewed in [1]. Machine vision has also been applied to optimize adaptive traffic-light control modes [2]. The objective of a Smart Traffic system is to combine real-time traffic sensing and intelligent decision making in order to improve traffic flow and reduce unnecessary delays.

II. RELATED WORK

Deep reinforcement learning has been studied as a data-driven approach for adaptive traffic signal control, and a comprehensive review of the area is provided in [1]. Machine-vision-based optimization of adaptive traffic-light control

modes was investigated in [2]. Both works show that traffic signals can use observed traffic state instead of fixed timers.

III. PROBLEM STATEMENT AND RESEARCH GAP

Conventional fixed-time systems assign predefined green, yellow, and red durations. However, traffic conditions vary with time of day, vehicle density, road capacity, weather, incidents, special events, and emergency situations. A traffic-control system should therefore observe current conditions and dynamically adapt signal timings.

A. Research Gap

Existing systems work well for regular daily traffic, but they lack a built-in way to automatically clear a path for a speeding ambulance during an emergency. The proposed work addresses this problem.

IV. OBJECTIVES

The objectives of the proposed Smart Traffic framework are:

- 1) To collect real-time traffic information.
- 2) To estimate traffic density.
- 3) To determine queue length and waiting time.
- 4) To dynamically adjust traffic signal timing.
- 5) To reduce unnecessary vehicle waiting.
- 6) To improve traffic flow.
- 7) To automatically clear a path for a speeding ambulance.
- 8) To provide a framework suitable for machine-learning and reinforcement-learning extensions.

V. COMPARISON OF TRAFFIC CONTROL SYSTEMS

Table I compares three approaches. Lower decision time means a faster decision.

TABLE I
COMPARISON OF TRAFFIC CONTROL SYSTEMS

System	Decision Speed (s)	Internet Use	Setup Cost (\$)
Basic Timed Lights	30	Zero	200
Cloud AI Systems	2	High	2,500
Local Smart Cameras	0.5	Low	1,200

VI. PROPOSED SMART TRAFFIC SYSTEM

The proposed system consists of five layers:

- 1) Traffic sensing
- 2) Communication
- 3) Data processing
- 4) Intelligent decision
- 5) Traffic signal control

The architecture is shown in Fig. 1.

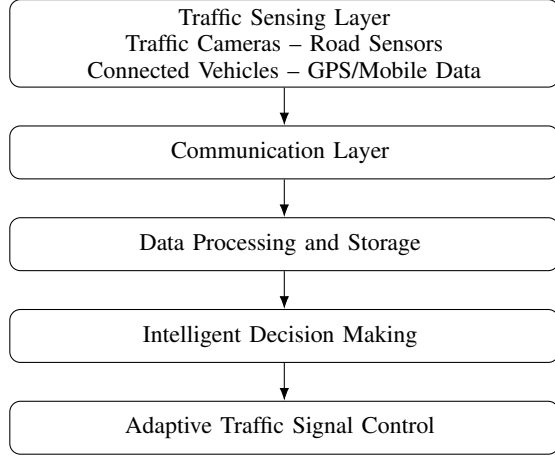


Fig. 1. Layered architecture of the proposed Smart Traffic system.

At deployment level, data moves in the fixed order shown in Fig. 2: street cameras send video to smart pole computers, which forward processed data to the city cloud server, which in turn controls the traffic lights.

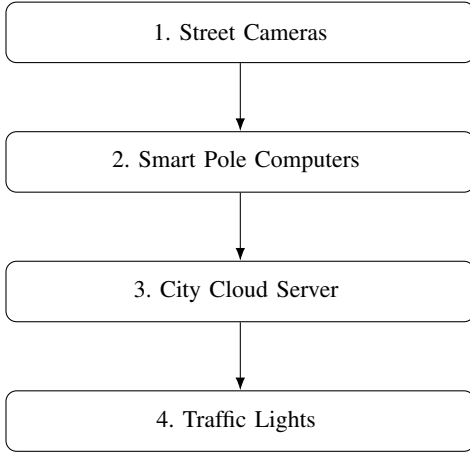


Fig. 2. Deployment architecture flow.

VII. SYSTEM WORKFLOW

The workflow of the control process is shown in Fig. 3. A conceptual intelligent intersection is shown in Fig. 4.

VIII. TRAFFIC DENSITY ESTIMATION

Traffic density is the number of vehicles occupying a specified road length:

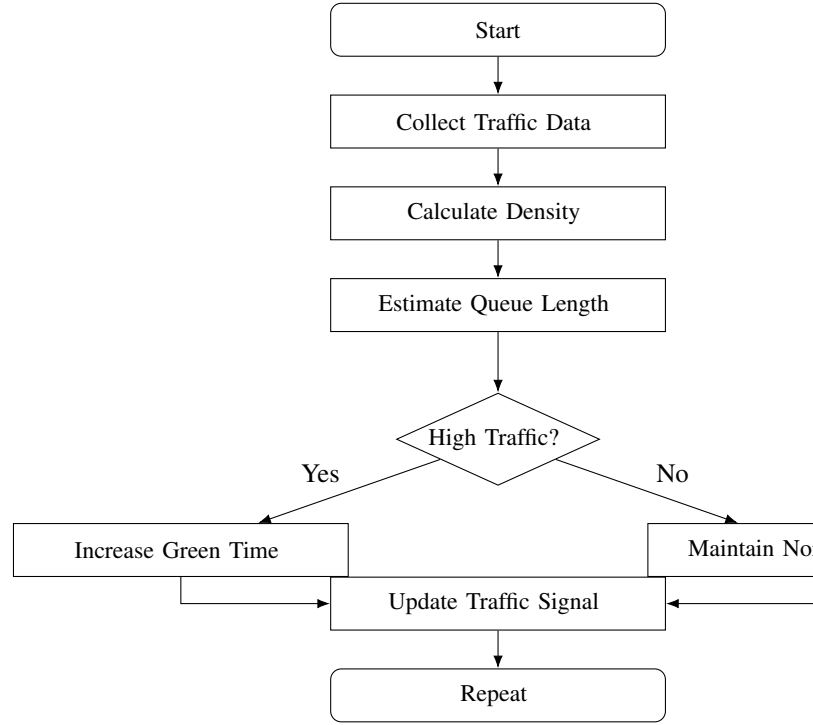


Fig. 3. Flowchart of the Smart Traffic control process.

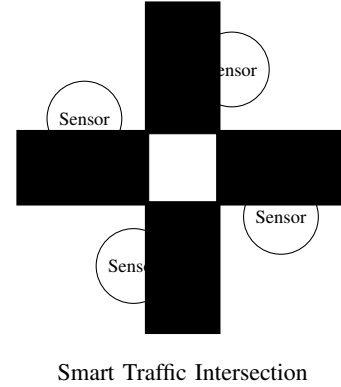


Fig. 4. Conceptual representation of a Smart Traffic intersection.

$$D = \frac{N}{L} \quad (1)$$

where D is traffic density (veh/km), N is the number of vehicles, and L is the road length (km).

IX. TRAFFIC FLOW

Traffic flow is the number of vehicles passing a location in a specified time:

$$Q = DV \quad (2)$$

where Q is traffic flow (veh/h), D is traffic density (veh/km), and V is average vehicle speed (km/h).

X. WAITING TIME

Average waiting time is an important performance measure for signal systems:

$$T_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N T_i \quad (3)$$

where T_i is the waiting time of vehicle i (s).

XI. TRAFFIC STATE AND DATA REPRESENTATION

The traffic state at an intersection is represented by the vector

$$S = [D_1, D_2, D_3, D_4, q_1, q_2, q_3, q_4]^T \quad (4)$$

where D_i is traffic density and q_i is queue length for approach i . Queue length uses lowercase q to avoid confusion with traffic flow Q in Eq. (2).

Traffic observations are collected in a matrix

$$X = [x_{jk}] \quad (5)$$

where each row is an observation and each column is a traffic parameter.

XII. ADAPTIVE SIGNAL CONTROL

The objective is to choose signal durations suited to traffic conditions:

$$J = w_1 T_{\text{avg}} + w_2 q_{\text{avg}} + w_3 D_{\text{avg}} \quad (6)$$

where w_1 , w_2 , and w_3 are weighting parameters.

The control action vector is

$$A = [G_1, G_2, G_3, G_4]^T \quad (7)$$

where G_i is the green-time allocation (s) for approach i .

When an ambulance is detected on approach k , the controller overrides Eq. (6) and sets G_k to green until the vehicle has cleared the intersection.

XIII. TRAFFIC DATASET

Table II presents an illustrative traffic dataset. Units are given in the column labels.

TABLE II
ILLUSTRATIVE TRAFFIC DATASET

Lane	Vehicles (count)	Density (veh/km)	Queue (vehicles)	Speed (km/h)
Lane 1	35	42	18	32
Lane 2	48	57	25	28
Lane 3	22	29	10	40
Lane 4	51	61	29	25

XIV. PERFORMANCE EVALUATION

Performance can be evaluated using waiting time, queue length, traffic flow, and adaptive control capability. The values in Table III are illustrative and must be replaced with actual experimental measurements before submission.

TABLE III
ILLUSTRATIVE PERFORMANCE METRICS

Metric	Conventional	Smart Traffic
Waiting Time (s)	45	30
Queue Length (vehicles)	32	21
Traffic Flow (veh/h)	650	820
Adaptive Control	No	Yes
Real-Time Data	Limited	Yes

XV. SMART TRAFFIC CONTROL PROCEDURE

The Smart Traffic control procedure consists of the following steps:

- 1) Initialize the traffic monitoring system.
- 2) Collect real-time traffic data.
- 3) Calculate vehicle density using Eq. (1).
- 4) Estimate the queue length.
- 5) Calculate average waiting time using Eq. (3).
- 6) Construct the traffic state vector (Eq. (4)).
- 7) Determine the control action (Eq. (7)), giving priority to any detected ambulance.
- 8) Update the traffic signal timing.
- 9) Monitor the resulting traffic condition.
- 10) Repeat the process continuously.

XVI. DISCUSSION

The framework provides a structure for integrating real-time sensing with adaptive signal control. Deep reinforcement learning is a potential approach for learning signal-control policies from traffic-state observations [1], and machine vision can supply the traffic data needed for adaptive control [2]. Local processing at the smart pole reduces internet use compared with cloud-only systems (Table I) while keeping decisions fast.

XVII. LIMITATIONS

The numerical values in Tables II and III are illustrative and do not come from a real deployment. The diagrams describe a conceptual architecture. Actual performance depends on sensors, communication infrastructure, data quality, controller design, and deployment conditions, so experimental validation is required before drawing conclusions.

XVIII. FUTURE WORK

Future research directions include:

- Multi-intersection traffic coordination.
- Real-time vehicle detection using computer vision.
- Deep reinforcement learning for adaptive signal control.
- Field testing of emergency vehicle priority.
- Connected and autonomous vehicles.
- IoT-based traffic monitoring.
- Edge computing for low-latency traffic decisions.
- Large-scale real-world deployment.

XIX. CONCLUSION

This paper presented a Smart Traffic framework for urban traffic management that integrates traffic sensing, data processing, traffic-state estimation, adaptive signal control, and performance evaluation, and that addresses the gap of automatically clearing a path for a speeding ambulance. Future research should validate the framework using real-world traffic datasets and experimental deployment.

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